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Palaeomagnetic study of Cretaceous red beds from the eastern Hindukush ranges, northern Pakistan: palaeoreconstruction of the Kohistan–Karakoram composite unit before the India–Asia collision

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SUMMARY

We report palaeomagnetic results obtained from Cretaceous red bed samples, collected from the eastern Hindukush mountains, northern Pakistan. Rock magnetic studies revealed specular haematite as the dominant remanence carrier, while pigmentary haematite, magnetite and goethite are responsible for the acquisition of secondary magnetizations. Thermal treatment generally revealed three components of magnetization. The most unstable component (A) was removed between 200 °C and 500 °C, which corresponds to the local present field direction. The intermediate component (B) was generally separated between 200 °C and 660 °C. The omnipresence of this complex secondary component across the Northern Suture Zone suggests that movement of chemically active orogenic fluids at, or slightly before, the time of the India–Asia collision was probably responsible for the authigenic remagnetization processes. The characteristic component (C) was generally unblocked between 600 °C and 685 °C. The characteristic directions from 12 sites fall into two groups. The sites from the Purit Formation give a tilt-corrected mean direction of $D = 311.2°$, $I = -4.4°$, $k = 12.4$, $\alpha_{95} = 17.8°$, indicating a counter-clockwise rotation of about 66° and 15° relative to Eurasia and India, respectively. The tilt-corrected mean direction for the Reshun and Drosh formations is $D = 263.0°$, $I = 2.2°$, $k = 52.7$, $\alpha_{95} = 10.6°$, which indicates that the sampling area experienced a counter-clockwise rotation of about 114° and 63° with respect to Eurasia and India, respectively. In both cases the palaeomagnetic rotation is consistent with the regional superstructure of the Hindukush–Pamir–Karakoram syntaxial bend. The difference in rotation between these two groups is probably due to local tectonic movement caused by the net rotation of thrust sheets along the Northern Suture Zone. The palaeolatitudes calculated for Purit and Reshun + Drosh formations are 2.2°S and 1.1°N, respectively. This indicates that about 4000 km of convergence has taken place since the mid-Cretaceous, of which about 90 per cent post-dates the India–Asia collision. The Late Cretaceous inclination values across the Northern Suture Zone are almost identical, which indicates that suturing had been completed at the time of ChRM acquisition.

Key words: Cretaceous, haematite, Hindukush, palaeolatitude, palaeomagnetism, tectonics.

INTRODUCTION

The Himalaya, Karakoram and Tibet are the most spectacular consequences of the collision of the Indian subcontinent with Eurasia in Late Cretaceous–Early Palaeogene time. An initial geological study by Tahirkheli (1979) in the northwest Himalaya has been followed by a large number of publications, bringing this previously unmapped and almost unknown region into the limelight. From these publications it is now well documented that the geology and tectonics of the northwest Himalaya are dominated by the Mesozoic island-arc sequence, termed the 'Kohistan–Ladakh arc sequence' (Tahirkheli 1979; Thakur 1983; Coward *et al.* 1986; Searle 1986; Windley 1983). The island-arc sequence separates the Karakoram microplate to the north from the Indian plate to the south. Until now, most of the geological and geophysical work has concentrated on the chronology of the arc development and the timings

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of suturing along the Main Karakoram Thrust (MKT) in the north and Main Mantle Thrust (MMT) in the south (Zeitler 1985; Pudsey 1986; Searle *et al.* 1987; Molnar 1988; Treloar *et al.* 1989). Palaeomagnetic data from the Indian subcontinent and southern Tibet show that, during the Cretaceous, Greater India occupied a southern hemisphere position with its northern margin at a palaeolatitude of about 20°S; however, no information has been available to determine the pre-Tertiary latitudinal position of the Kohistan and Karakoram terranes. Geological evidence suggests that most of the sedimentary and volcanic rocks in the Himalaya–Karakoram collision belt are highly metamorphosed, so it is extremely difficult to obtained pre-collision palaeomagnetic directions (especially from magnetite-bearing rocks). The available characteristic remanent magnetization (ChRM) directions from the eastern Hindukush and Ladakh regions have already been interpreted as a secondary remagnetization, attributed to the India–Asia collision (Klootwijk & Conaghan 1979; Klootwijk *et al.* 1979, 1994).

In the last decade, palaeomagnetic studies have been made on red beds of various geological ages from the central Asian terranes, which span from the Tajik depression in the west and the Tibetan Plateau to the east (Bazhenov 1993; Bazhenov *et al.* 1994; Pozzi & Feinberg 1991; Thomas *et al.* 1993; Chen *et al.* 1993a; Frost *et al.* 1995). On the basis of palaeomagnetic results, several workers have described the geographical and tectonic evolution of the lithospheric blocks that have accreted to form the central Asian mobile belt (Enkin *et al.* 1992; Chen *et al.* 1993b; Shangyou *et al.* 1990). In the Cretaceous palaeogeographical reconstruction of Asia (Chen *et al.* 1993b; Fig. 8), there is an unclear Neo-Tethyan Ocean gap roughly at the longitude of the Kohistan–Karakoram–Pamir area. Keeping in view these problems, we propose a new interpretation for the evolution of this complex region, with the presentation of palaeomagnetic results from the least deformed Cretaceous red beds of the Kohistan and Karakoram terranes. The purpose is to highlight the Cretaceous palaeogeography and post-Cretaceous tectonic activity around the northwestern Himalaya, with special reference to the India–Asia collision.

REGIONAL TECTONICS

Several workers have speculated that prior to the India–Asia collision and from the Early Mesozoic onwards many small microcontinents were separated from the northern margin of Gondwana and accreted to stable Eurasia. Stöcklin (1977) predicted that the broad arcuate belt between the Axial Ophiolite Belt in the south and the Hindukush–Wanch–Akbaytal Fault lineament in the north constitutes an amalgamation of microcontinental blocks of Gondwanic origin (see the regional tectonic setting in Fig. 14). Tapponier *et al.* (1981) marked the northern boundary of the Gondwanic continents to lie along the Herat and Panjshir faults in Afghanistan, to swing around the central Pamir and then along the western Altyn Tagh and Kunlun faults in Tibet. They further suggested that nearly all of the present curvature of the Pamir Syntaxis would have been caused by the wedging of India's northwestern tip into Asia.

Magnetic anomaly data from the Indian Ocean suggest that the separation of India from Australia and Antarctica started at about 130 Ma (Molnar & Tapponier 1975; Powell 1979). The spreading rate was 10–20 cm yr^{-1} between 80 and 50 Ma, but immediately after collision the rate of spreading decreased to about 5 cm yr^{-1}. In latest Oligocene–earliest Miocene, underthrusting of the Indian plate beneath Asia/or Gondwanic microcontinents was initiated in the Himalaya (Besse *et al.* 1984; Patriat & Achache 1984). On the basis of palaeomagnetic data, Klootwijk *et al.* (1992, 1994) estimated the initial collision between India and the Kohistan–Karakoram composite unit at or before the Cretaceous–Tertiary boundary (KTB).

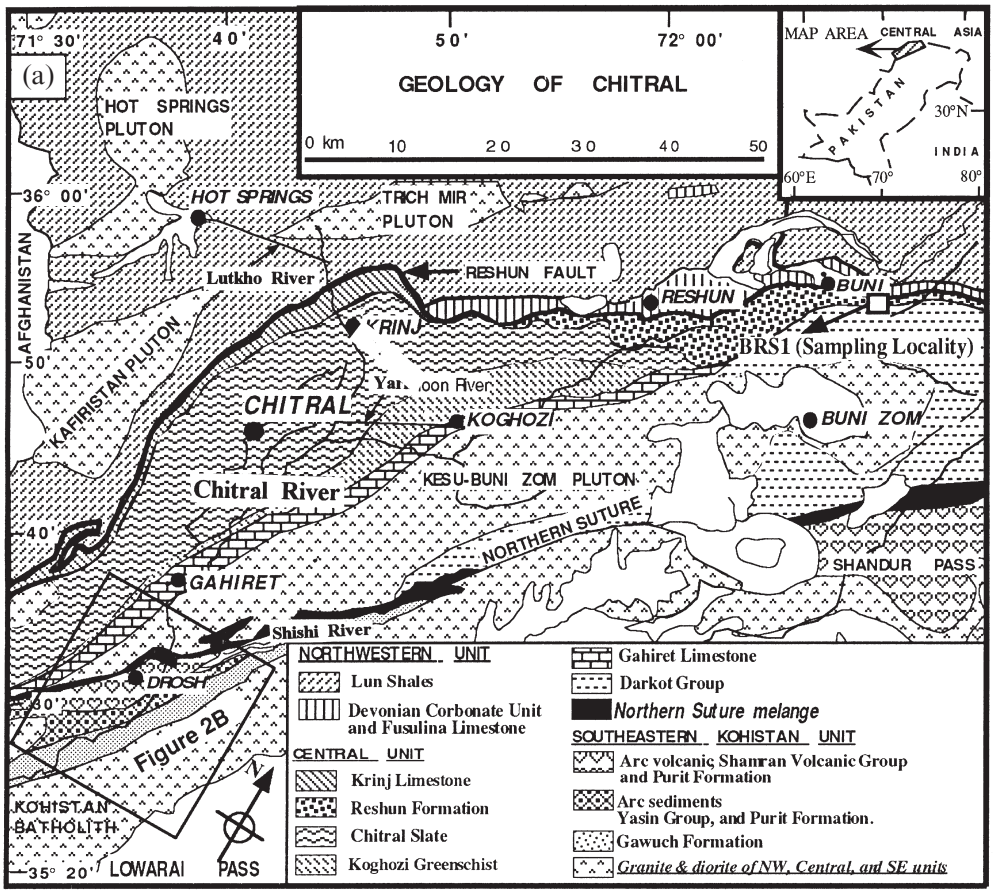
At present, northwestern Pakistan contains the junction of the world's three biggest mountain ranges: the Himalaya, Karakoram and Hindukush, which result from the India–Asia collision. The Indus Suture Zone (or Indus–Zangpo or Yarlung–Tsangpo in Tibet), the crustal lineament separating the Indian plate from the South Asian margin, splits into two branches at the northwestern Himalayan Syntaxis (Fig. 14). The northern branch of the suture zone (MKT by Tahirkheli 1982 and Northern Suture by Pudsey 1986) separates the Palaeozoic to Mesozoic Tethyan shelf sediments of the South Asian margin to the north from the Jurassic to Cretaceous Kohistan–Ladakh arc sequence to the south (called the Northern Suture Zone 'NSZ' in this study). The southern branch of the suture zone (MMT in Pakistan by Tahirkheli *et al.* 1979 and Indus Suture in Ladakh by Searle 1983) separates the Kohistan–Ladakh island arcs from the Precambrian to Mesozoic gneisses, sediments or volcanics of the Indian plate to the south. The Kohistan–Ladakh arc sequence is divided into two parts by a projection of Indian basement terrane known as the Nanga Parbat–Haramosh massif, with the Kohistan domain to the northwest of the massif and the Ladakh domain to the southeast. The Kohistan–Ladakh island arcs were accreted onto the Karakoram microplate during the Cretaceous when the back-arc basin between the two terranes closed, forming the NSZ (Pudsey 1986). This event has been placed between 102 and 85 Ma from ^{39}Ar–^{40}Ar and K–Ar dating studies and cooling histories (Treloar *et al.* 1989). With the closure of the NSZ, the Kohistan–Ladakh terrane formed the southern margin of Asia. The Aptian–Albian Yasin Group Limestones along the northern margin of Kohistan (and their westward extension in the Drosh–Shishi area) represent the youngest marine sedimentation before the accretion of the island-arc sequence onto the Karakoram microplate.

GEOLOGY OF THE SAMPLING AREA

The sampling area is situated on the southern flanks of the Hindukush mountains, which tectonically forms part of the Kohistan island arc and the Karakoram microplate. On the basis of reconstructed stratigraphical sequences and structural evolution, Pudsey *et al.* (1985) divided the area into the following three tectonic units (Fig. 1a).

(1) The northwestern or Lutkho–Turikho unit occupies the area between the Pakistan–Afghanistan border and the north-dipping Reshun Fault. This unit comprises Devonian carbonates and quartzites which are overlain by Devonian to Permian shales (Lun shales) and slates with some limestones.

(2) The central or Kunar–Yarkhun unit forms the area between the Reshun Fault and the NSZ. In addition to the red beds of the Reshun Formation (RF), the unit includes north-dipping Upper Palaeozoic slates and quartzites (Darkot



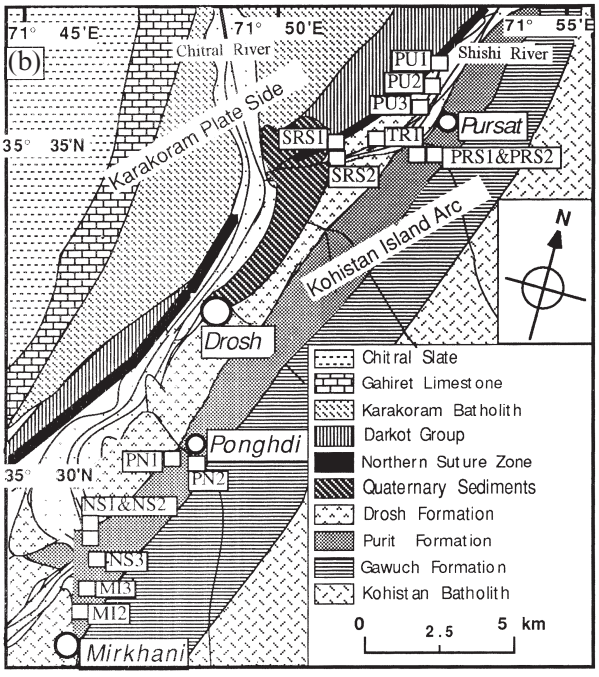


Figure 1. Schematic geological maps of the study area (modified from Calkins *et al.* 1981; Pudsey *et al.* 1985) with palaeomagnetic sampling sites. (a) Schematic geological map of the Chitral district. The northwestern unit represents the area between the Reshun Fault and the Pakistan–Afghanistan border. The central unit forms the area between the Reshun Fault and the Northern Suture Zone. The southeastern Kohistan unit is located between the Northern Suture Zone and the Main Mantle Thrust. As shown in the map, one locality, BRS1, was sampled from the Reshun Formation of the central unit, which lies on the Karakoram plate side. (b) Schematic geological map of the Drosh–Shishi area within the southeastern Kohistan unit, where most of the palaeomagnetic samples were collected.

Group), probably faulted against the Chitral Slates, Koghozi Greenschists and Cretaceous Krinj and Gahiret Limestones. Structurally, the central unit forms an elongate synclinal belt, which follows the Chitral and Mastuj Rivers, with the Chitral Slates in the central part of the syncline (Calkins *et al.* 1981).

(3) The southeastern or Kohistan unit is placed between the NSZ and the MMT. It comprises Cretaceous volcanic rocks with some sediments intruded by diorites, tonalites and granites. The northwestern and central units represent the southern margin of Asia, which are intruded by granitic and granodioritic plutons of the Western Karakoram Batholith. The Northern Suture is a zone of mélange, 150 m to 4 km wide, which contains blocks of volcanic greenstones, limestones, red shales, conglomerates, quartzites and serpentinites in a matrix of strongly cleaved slate with some coarser clastic interbeds (Pudsey 1986). The mélange is interpreted by Pudsey (1986) as an olistostrome, largely derived from the Kohistan arc to the south.

As shown in Figs 1(a) and (b), all palaeomagnetic samples were collected from red bed rocks in the Buni and Drosh–Shishi areas. The red beds from the RF include polymict pebble and cobble conglomerates, red calcareous shales and pale grey micritic limestones lie below the Reshun Fault and above Chitral Slates and/or the Darkot Group of sediments (with total thickness varying from 140 m to nearly 1500 m). The conglomerate beds are thick and lenticular, with some intervening cross-bedded sandstones. Some of the pebbles in the limestone breccia and conglomerates indicate an Aptian–Albian age (Talent *et al.* 1982; Pudsey *et al.* 1985). The age of the conglomerate beds is therefore younger than Aptian–Albian and could be Late Cretaceous. The rocks of the RF (least deformed unit of fluvial origin) were deposited just after the Kohistan–Karakoram collision along the NSZ. Its petrography suggests that it was derived largely from the volcanic arc (Pudsey 1986).

The Cretaceous rocks in the Shishi valley are divided into three different rock units (Pudsey *et al.* 1985).

(1) The basal unit is the Gawuch Formation (GF), which predominantly consists of green to dark green phyllites that are strongly cleaved and lineated. The limestones in this formation indicate a marine environment.

(2) The overlying unit of the Purit Formation (PF), about 1 km thick in the southwest and wedging out half-way along the Shishi valley, has been interpreted as fluvial (Pudsey *et al.* 1985). Red calcareous shale with abundant calcite veins is the dominant rock type. Quartz and feldspar are present in widely varying proportions, with magnesian chlorite, illite and haematite. In the southwest, conglomerate and sandstone constitute about one-third of the formation.

(3) The next overlying unit is the Drosh Formation (DF), which is a sequence of thickly bedded porphyritic andesites. Some thin red calcareous shales are interbedded with the lava flows. Chalcopyrite is an accessory mineral in some flows (Calkins *et al.* 1981). A belt of limestones and phyllites overlies the volcanic greenstone of the DF. The unit consists mainly of dark-grey, thin bedded limestones interbedded with black phyllites and black calcareous shales. *Orbitolina*, found in calcareous limestone beds on the south of Drosh, indicates a Cretaceous age for this unit (Calkins *et al.* 1981). Pudsey *et al.* (1985) included this unit as part of the Northern Suture mélange.

Tectonically this narrow belt of Cretaceous rocks in the Drosh–Shishi area lies between the Naz Bar and Shishi faults. The rocks in this zone are only slightly metamorphosed, and well-documented fossils from the northeastern part of the succession confirm the Cretaceous age of these rocks (Aptian–Albian: Pudsey *et al.* 1985). Bedding as well as foliation dips steeply to the west.

We collected more than 350 palaeomagnetic samples of Devonian to Cretaceous sedimentary, volcanic and plutonic rocks from the northwestern, central and southeastern (Kohistan) units. Here we only report on palaeomagnetic results obtained from Cretaceous red beds of the Buni and Drosh–Shishi areas (shown in Figs 1a and b, and listed in Tables 1 to 3).

FIELD AND LABORATORY PROCEDURE

Palaeomagnetic samples of Cretaceous red beds were collected from four different lithostratigraphic units. Eight to 14 block samples were collected from each site using a tripod magnetic compass (dip-direction method). Most of the block samples were cored at the newly established Geoscience Laboratory, Islamabad. Two to five specimens of 2.5 cm in diameter and 2.2 cm in height were cored from each block sample.

The palaeomagnetic measurements were made at Kyoto University, Japan, using a two-axis cryogenic magnetometer (ScT C-112). Since alternating field demagnetization proved to be ineffective, most samples were thermally demagnetized using a non-inductive electric oven with less than 5 nT residual field. For the reliable isolation of natural remanent magnetization (NRM) components, 13–24 heating steps were made. Above 660 °C, most of the specimens were heated at 3–5 °C, in order to separate the high-temperature components in a reliable way. Low-field magnetic susceptibility was measured after each heating step using a Bartington M.S.2 susceptibility meter, to detect mineralogical changes during heating. NRM components were obtained by principal component analysis (Kirschvink 1980). Fisher statistics were used to calculate the mean direction (Fisher 1953)

Thermomagnetic analyses were performed for Curie temperature determination using an automatic recording semi-horizontal magnetic balance. Isothermal remanent magnetization (IRM) acquisition and back-field were applied to study the coercivity of the magnetic minerals. Three orthogonally composite IRMs of 1.3, 0.4 and 0.12 T were applied as hard, medium and soft coercivity fractions, and then thermally demagnetized for distinct unblocking temperatures (Lowrie 1990). An IRM of 1 T was imparted at 5 K and the sample was continuously heated up to 300 K using a Quantum Design Magnetic Property Measurements System (MPMS) to investigate the low-temperature magnetic phase transition.

ROCK MAGNETISM

The rock magnetic properties were investigated for typical samples from all red bed formations. The various formations have quite distinct magnetic mineralogies.

The Reshun Formation

The sample from very coarse-grained sandstone (BRS1-Q1.2) shows almost no thermal enhancement in magnetic susceptibility when measured at room temperature, while the sample from

Table 1. Sample detail of the Reshun, Drosh, and Purit formations and the Northern Suture Zone.

Name of the formation	*Lithology*	*Maximum stratigraphic thickness*	*Site name (No. of samples collected)*	*Bedding attitude (strike/dip in degrees)*	*Site location (latitude and longitude)*
Reshun Formation (RF)	Red shales, grey micritic limestone, and pebble to cobble conglomerates	1.5 km	BRS1 (17)	71/70	North side of Buni town, on the Mastuj road (36°16′33″N, 72°19′41″E)
Northern Suture Zone (NSZ)	Thin-bedded limestone with black calcareous shale	150 m to 4 km wide	SRS1 (12) SRS2 (12)	48/81 25/56	Near Shishi village in the Shishi valley (35°36′7″N, 71°50′41″E)
Drosh Formation (DF)	Thin layers of red shales interbedded with andesitic lavas	1.5 km	PU1 (13) PU2 (10) PU3 (13)	1/48 6/44 10/32	Across Purgal village in the shishi valley (35°36′26″N, 71°51′52″E)
			PN1 (8)	30/45	Near Ponghdi village (35°30′34″N, 71°47′12″E)
Purit Formation (PF)	Red calcareous shales, lithic to feldspathic sandstone, and conglomerates	1 km	PRS1 (11) PRS2 (9)	44/56 50/83	Near Pursat village in Shishi valley (35°35′53″N, 71°51′24″E)
			NS1 (9) NS2 (11) NS3 (8)	69/43 68/40 39/69	Across the Nagar Fort on the main Chitral road (35°29′11″N, 71°44′43″E)
			MI2 (10) MI3 (10)	20/40 35/79	Near Mirkhani post on the main Chitral road (35°28′9″N, 71°44′6″E)

Table 2. Site-mean and formation-mean directions of the intermediate component (B).

Formation	*Site code*	*N/n*	*In geographic coordinates*				*In stratigraphic coordinates*			
			Dec.(°)	*Inc.(°)*	*k*	α_{95}(°)	*Dec.(°)*	*Inc.(°)*	*k*	α_{95}(°)
Reshun (RF)	BRS1	6	136.1	-48	19.3	15.7	144.2	+17.6	19.3	15.6
Purit (PF)	PRS1	8	43.3	-44.8	4.1	31.5	93.0	-21.1	5.3	2.6
	PRS2	6	70.3	-79.0	11.2	20.9	129.7	-3.0	11.2	20.9
	NS3	4	308.3	+86.0	14.8	24.7	308.9	+17.0	14.8	24.7
	MI2	5	247.8	+42.3	6.1	33.8	260.8	+9.5	6.6	32.3
	MI3	7	194.7	+77.9	7.1	24.2	293.3	+15.0	7.2	24.2
Drosh (DF)	PU1	10	87.0	-31.9	53.4	6.7	88.9	+15.7	53.4	6.7
	PU2	6	92.0	-41.4	91.3	7.0	93.0	+2.5	91.1	7.1
	PU3	5	108.5	-27.1	20.6	17.3	109.6	+4.3	18.5	18.3
Formation-mean direction		9^n	266.3	+57.0	8.2	19.1	288.9	3.0	10.2	17.0

N = number of samples used for site-mean calculation; n = number of sites used for formation-mean calculation. Dec. = declination; Inc. = inclination; k = estimate of precision parameter; α_{95} = semi-angle of the circle of 95% confidence about the mean direction.

Table 3. Site-mean and formation-mean directions for the high-temperature component (C).

Formation	*Site code*	*N/n*	*In geographic coordinates*				*In stratigraphic coordinates*			
			Dec. (°)	*Inc. (°)*	*k*	α_{95}(°)	*Dec. (°)*	*Inc. (°)*	*k*	α_{95}(°)
Reshun (RF)	BRS1	6	68.2	-15.9	10.8	21.4	85.2	-7.9	10.8	21.3
Purit (PF)	PRS1	8	128.4	-77.9	9.8	18.7	137.9	-12.0	18.6	13.2
	PRS2	7	86.2	-76.2	54.4	8.3	129.1	+1.2	54.8	8.2
	NS1	4	322.9	+16.0	10.3	30.0	321.9	-25.3	10.3	30.1
	NS2	5	297.7	-2.1	6.0	34.1	287.5	-32.6	6.0	34.1
	NS3	6	15.3	+73.6	68.1	8.2	324.4	+13.7	67.7	8.2
	MI2	5	298.1	+50.9	29.6	14.3	295.2	+11.1	29.6	14.3
	MI3	7	341.9	+65.4	6.6	25.4	319.7	-8.8	6.6	25.4
Drosh (DF)	PU1	10	74.5	-32.0	128.2	4.3	76.6	+14.5	127.9	4.3
	PU2	6	67.6	-49.1	165.1	5.2	77.6	-8.2	165.2	5.2
	PU3	6	83.4	-39.1	16.8	16.8	87.0	-8.1	16.8	16.8
	PN1	5	258.2	+37.8	3.8	45.0	268.3	+1.0	3.8	45.0
Northern Suture	SRS1*	10	14.6	+36.6	45.3	7.3	3.5	-20.1	45.4	7.2
Zone (NSZ)	SRS2*	6	357.3	+34.6	135.5	5.8	341.7	+0.1	135.5	5.8
Formation-mean direction	PF sites	7^n	313.0	+55.0	5.5	28.4	311.2	-4.4	12.4	17.8
Formation-mean direction	RF+DF sites	5^n	74.3	-35.0	37.1	12.7	83.0	-2.2	52.7	10.6
Antipodal direction	RF+DF sites	5^n	254.3	35.0	37.1	12.7	263.0	2.2	52.7	10.6

Number of sites/samples/specimen used for Formation mean directions

Purit Formation.....7 sites/42 samples/64 specimen

Reshun+Drosh formations.....5 sites/33 samples/40 specimen

*Not used for mean calculation (see table 2 for further explanations)

fine-grained mudstone (BRS1-M2.2) has the highest increase after heating up to 600 °C (Fig. 2a). This high increase indicates the production of a ferrimagnetic impurity, most likely magnetite (Dekkers & Linssen 1989), which is more pronounced in the finer-grained sample than in the coarser one. The production of magnetite by heating red beds is probably due to the breakdown of iron silicates (Dunlop 1972). Thermomagnetic analysis of the mudstone samples shows a Curie temperature of about 680 °C (Fig. 2b). The IRM acquisition and back-field curve of the coarse-grained sample shows a dominance of high-coercivity magnetic minerals, while the presence of a low-coercivity mineral is also indicated below 200 mT (Fig. 2c). Thermal demagnetization of the composite IRMs (Lowrie 1990) shows a variation in magnetic minerals with lithology: in the mudstone sample a small amount of magnetite and goethite is associated with the dominant haematite (Zaman & Torii 1997), while a sample from coarse-grained sandstone (BRS1-Q2.1) shows the dominant presence of goethite in association with magnetite and haematite (Fig. 2d). According to optical microscopic observations (Zaman & Torii 1997), the dominant magnetic mineral identified in the mudstone sample is detrital specular haematite (haematite–ilmenite intergrowth), with pigmentary haematite present as subordinate magnetic mineral. In coarse-grained sandstone, goethite is observed as the dominant magnetic mineral, while haematite and magnetite are present as subordinate magnetic minerals.

The Purit Formation

Thermomagnetic behaviour measured in air reveals a Curie temperature of about 680 °C, indicating that the magnetic carrier is predominantly haematite (Fig. 3a). Thermal demagnetization of the composite IRMs shows the dominance of high-stability magnetic minerals (Fig. 3b). In the low-temperature curve

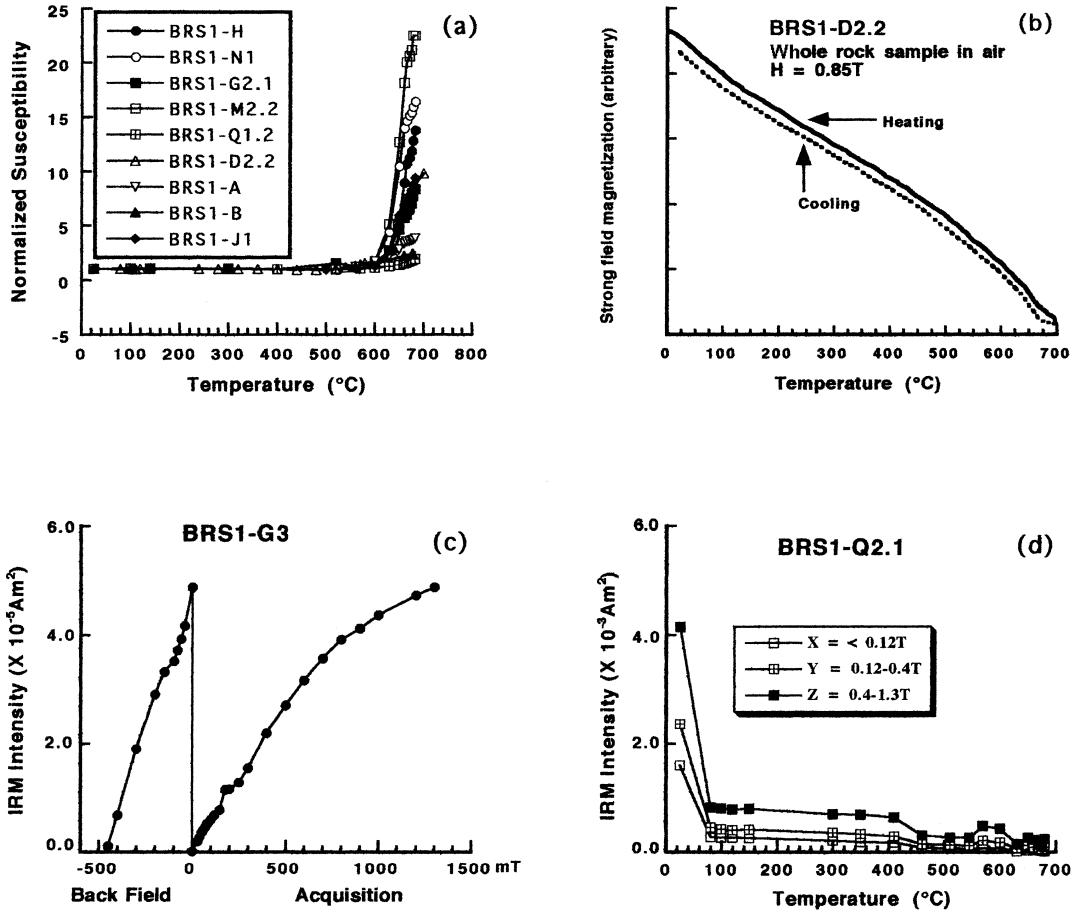


Figure 2. Results of the rock magnetic study for samples from the Reshun Formation, Buni locality. (a) Low-field magnetic susceptibility measured at room temperature after each heating step. (b) Strong-field thermomagnetic analysis of the whole rock sample in air. (c) IRM acquisition and backfield plot for coercivity spectrum analysis. (d) Thermal demagnetization of a three-components IRM produced by magnetizing the sample in successive fields of 1.3, 0.4 and 0.12 T along three different axes (Lowrie 1990).

(Fig. 3c), the magnetization phase around 250 K may correspond to the Morin transition of haematite, while a small phase transition at around 120 K indicates the minor presence of magnetite. From optical microscopic studies it is observed (Zaman & Torii 1997) that detrital specular haematite (haematite–ilmenite intergrowth) is the dominant magnetic carrier, while fine-grained pigmentary haematite is present as subordinate magnetic mineral.

The Drosh Formation

The curve of the thermomagnetic analysis shows two phases of magnetic minerals, one with a Curie temperature of about 575 °C and the other at 680 °C, indicating magnetite and haematite, respectively (Fig. 4a). The IRM acquisition and back-field curves indicate two phases of magnetic minerals (Fig. 4b). The low-coercivity phase (below 200 mT) may indicate the presence of magnetite, and the unsaturated phase is clearly related to haematite. The low-temperature curve also represents two magnetic mineral phases (Fig. 4c). A sharp Verwey transition around 125 K indicates the presence of magnetite, whereas the change around 240 K is probably related to the Morin transition of haematite.

PALAEOMAGNETISM

The Reshun Formation

Buni locality

Here we collected 17 orientated block samples from the red bed rocks, including mudstone, siltstone and coarse-grained sandstone at one site (BRS1). Bedding attitude was observed with alternate bedding of silty and sandy parts of the outcrop. The initial NRM intensities are in the range 6.67×10^{-04}–4.67×10^{-03} A m^{-1}. Three to four remanent components were determined on the basis of principal component analysis (Figs 5a and b). The remanent component that was removed below 300 °C is parallel to the present local field direction and hereafter called the A component. Two intermediate thermal stability components (B1 and B2) were isolated from 300 °C to 560 °C, and 560 °C to 660 °C, respectively (Fig. 5a). The high-temperature C component was unblocked between 660 °C and 678 °C. In some samples, the C component was broken down between 600 °C and 680 °C (Fig. 5b). Comparison with the rock magnetic results implies that the C, B1, and B2 components are carried by primary haematite, pigmentary haematite, and magnetite, respectively.

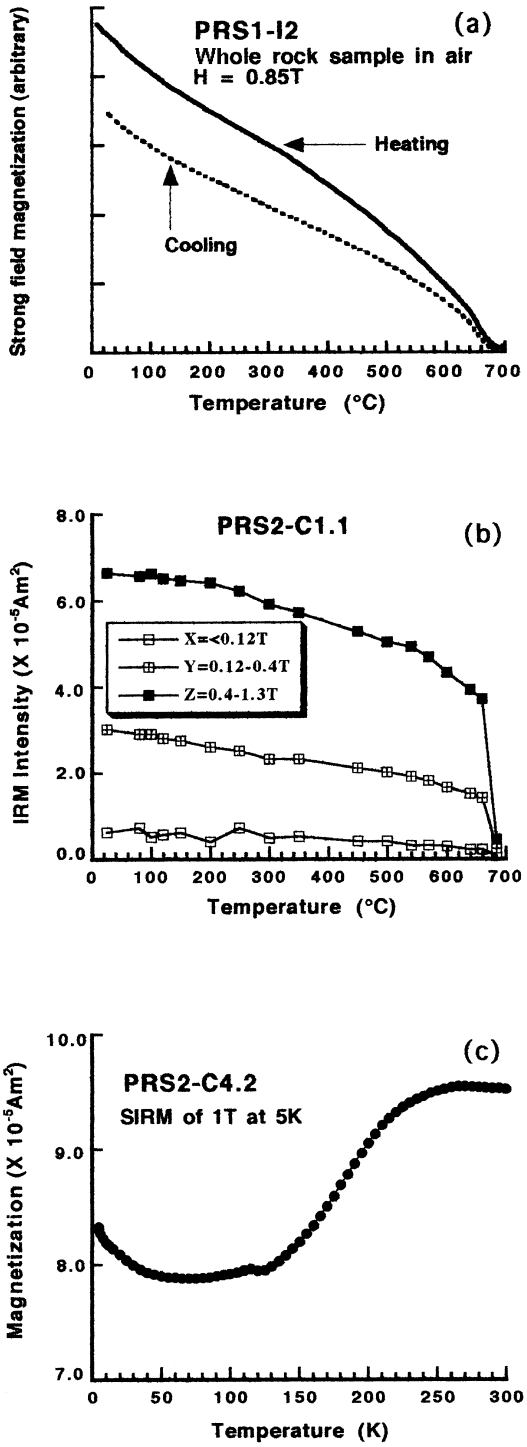


Figure 3. Results of the rock magnetic study for samples from the Purit Formation, Pursat locality. (a) Strong-field thermomagnetic analysis in air. (b) Thermal demagnetization of a three-components IRM produced by magnetizing the sample in successive fields of 1.3, 0.4 and 0.12 T along three different axes. (c) A saturation isothermal remanent magnetization (SIRM) of 1 T was imparted at 5 K and the sample was then heated up to 300 K for the determination of the low-temperature transitions of the magnetic minerals.

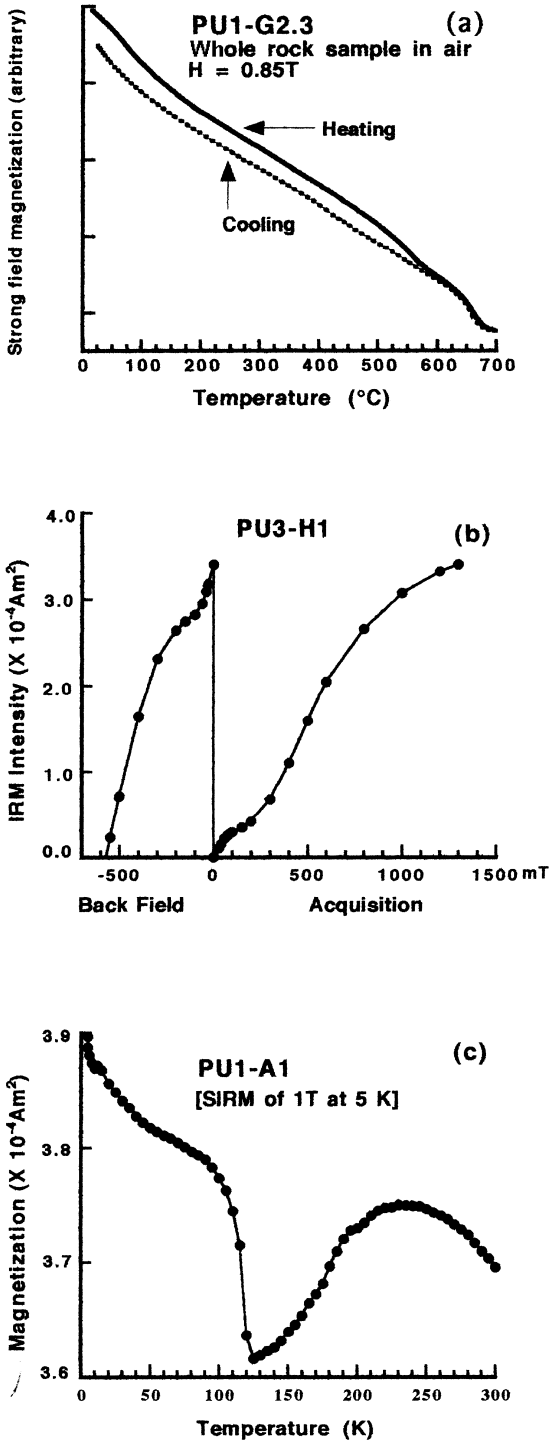


Figure 4. Results of the rock magnetic study for samples from the Drosh Formation, Purgal locality. Comments as for Fig. 3 except for (b), which is the IRM acquisition and backfield plot for the coercivity spectrum analysis.

The Purit Formation

Pursat locality

We collected 20 block samples from the red shales and red sandstone at two sites (PRS1 and PRS2). The initial NRM intensities varied between 5.26×10^{-04} and 3.53×10^{-03} A m^{-1}.

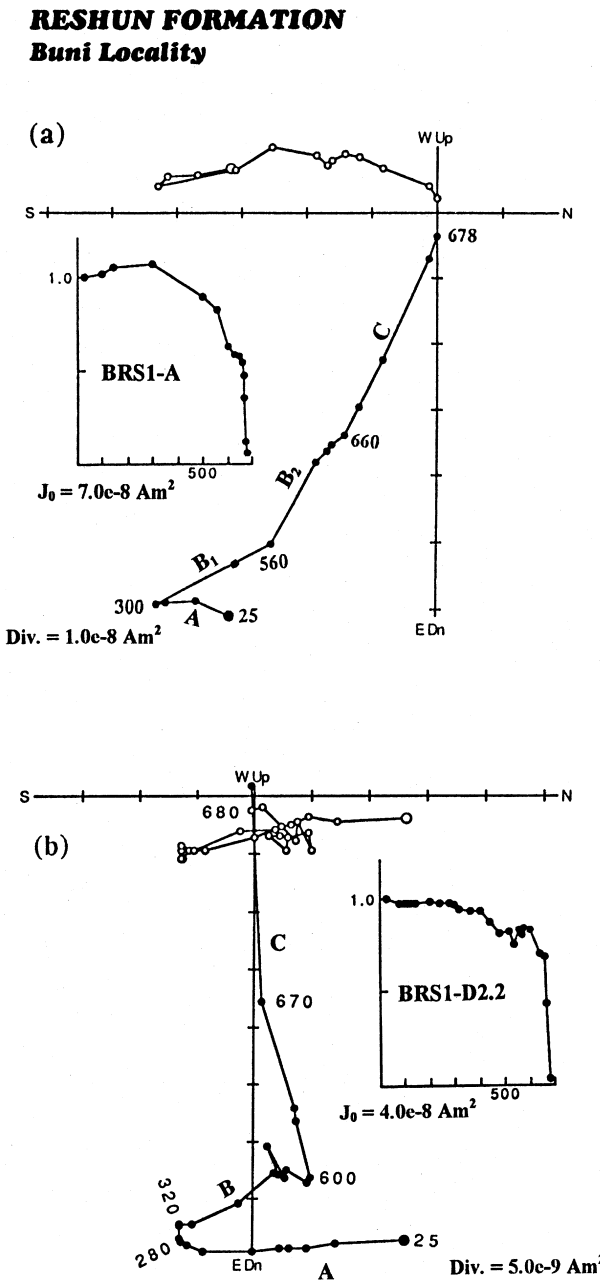


Figure 5. Results of the palaeomagnetic study for samples from the Reshun Formation at the Buni locality. (a), (b) Orthogonal projections and intensity plots of thermal demagnetization in stratigraphic coordinates. Solid symbols indicate projection onto the horizontal plane, and open symbols indicate projection onto the vertical plane. Numbers beside the projection indicate peak demagnetization temperatures in degrees Celsius. The components of magnetization in (a) and (b) are marked by A, B_1, B_2, C and A, B, C, respectively.

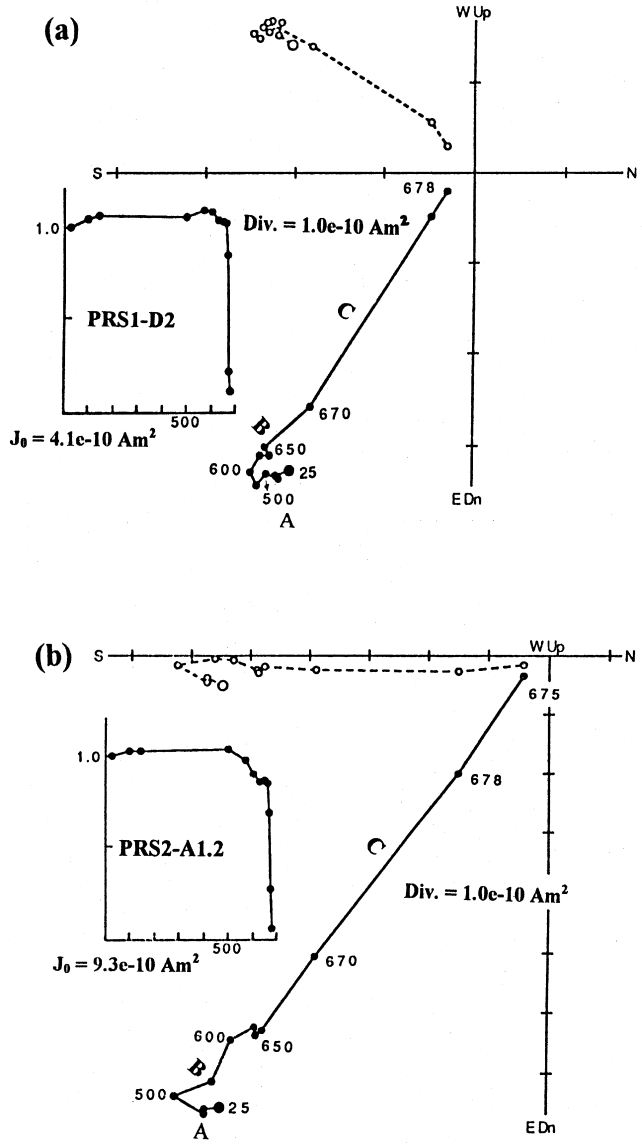


Figure 6. Results of the palaeomagnetic study for samples from the Purit Formation at the Pursat locality. (a), (b) Comments as for Fig. 5(b).

Three magnetization components were separated (Figs 6a and b). The low-temperature component (A) corresponds to the recent field direction, and was removed by heating up to 500 °C. An intermediate component (B) was generally unblocked between 500 °C and 670 °C. The most stable C component was isolated by heating up to 678 °C. The intensity of the remanence is dominantly held by the C component (Figs 6a and b). Comparison with the rock magnetic and optical microscopic results implies that secondary (pigmentary) and primary haematite phases are the carriers of the B and C components, respectively.

Nagar Fort locality

We collected 28 block samples from fine-grained red mudstone and siltstone at three sites (NS1, NS2, and NS3). The initial NRM intensities varied between 2.8×10^{-04} and 7.33×10^{-03} A m^{-1}. As for the Pursat locality, three components of magnetization were separated (Figs 7a,b and c). A recent field component (A) was generally removed below 450 °C. As shown in Figs 7(a) and (c), the B component was generally unblocked between 450 °C and 665 °C, whereas the

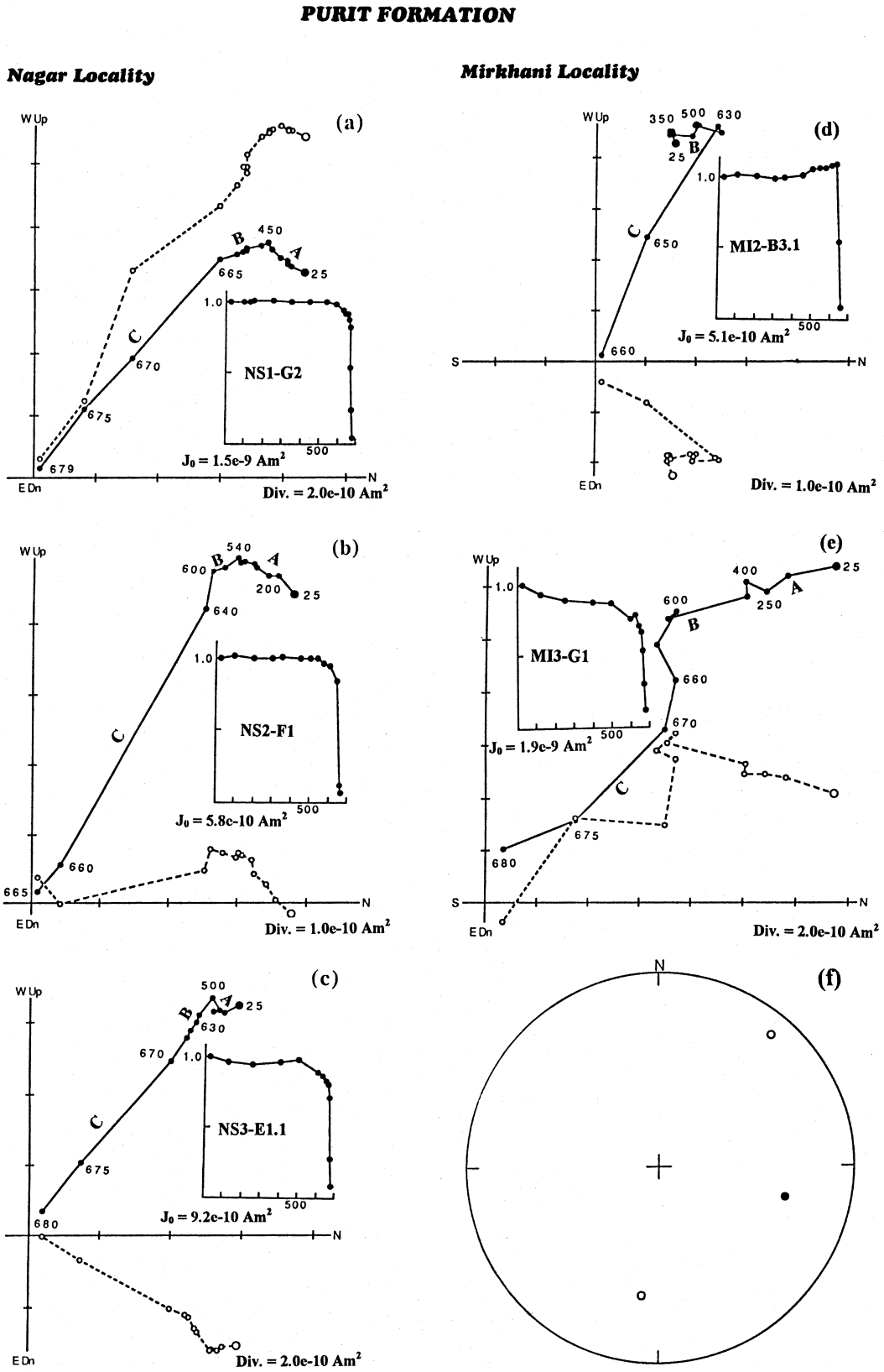


Figure 7. Results of the palaeomagnetic study for samples from the Purit Formation at the Nagar and Mirkhani localities. (a), (b), (c), (d), (e) Comments as for Fig. 5(b). (f) Randomly distributed directions of three conglomerate samples from the Mirkhani locality indicate a positive conglomerate test.

C component was isolated between 665 °C and 680 °C. In a few samples, the B and C components were isolated from 500 °C to 600 °C and 640 °C to 665 °C, respectively (Fig. 7b). The NRM intensity is dominantly carried by the characteristic component (Figs 7a,b and c). Comparison with the magnetic mineralogy implies that B and C components are carried by specular and pigmentary haematite, respectively.

Mirkhani locality

We collected 20 orientated block samples from the red sandstones at two sites (MI2 and MI3). Because of unclear local bedding, the attitude of an adjacent outcrop was observed. The NRM intensities measured at the initial stage varied widely between 7.33×10^{-04} and 1×10^{-02} A m^{-1}. Progressive thermal demagnetization generally revealed three components of magnetization (Figs 7d and e). A low-temperature component (A) was removed by heating up to 350 °C and is parallel to the present field direction. The intermediate component (B) was generally broken down between 350 °C and 630 °C. The characteristic component (C) in the MI2 samples was removed between 640 °C and 660 °C (Fig. 7d), and further heating resulted in a very noisy behaviour. Seen in conjunction with the relatively low coercivity values of the same sample, this type of behaviour indicates that the C component is carried by the coarse-grained haematite. In the case of MI3, the C component was unblocked between 660 °C and 680 °C (Fig. 7e). This and the optical microscopic observations suggest that pigmentary haematite and goethite are probably responsible for the acquisition of the B and A components, respectively.

To evaluate the stability of the ChRM component in the PF rocks, we collected four clast samples from a conglomerate bed, which is interbedded with the red sandstone at the Mirkhani locality and thus is synchronous. The ChRM obtained from three clast specimens shows a random distribution, indicating a positive conglomerate test (Fig. 7f). The positive test from an intraformational conglomerate provides very strong evidence that the ChRM in the PF samples is of primary origin.

The Drosh Formation

Purgal locality

Here 36 orientated block samples were collected from dark-reddish shale at three sites (PU1, PU2, and PU3). The initial NRM intensities were much higher than those of the PF and RF samples, and varied between 5.33×10^{-03} and 1.86×10^{-02} A m^{-1}. Generally, three remanent components were determined (Figs 8a,b and c). The low-temperature component (A) was generally removed by heating up to 300 °C. The B component was isolated, in general, between 200 °C and 570 °C. The C component was demagnetized by heating up to 680 °C. The intensity plots show thermally distributed blocking-temperature spectra, which indicate that more than half of the NRM intensity is carried by the C component and the remaining part is shared by the A and B components (Figs 8a,b and c). This and the the rock magnetic results imply that the C and B components are carried by haematite and magnetite, respectively.

Ponghdi locality

This locality has mainly red mudstone and coarse-grained siltstone. We collected eight orientated block samples from the mudstone and siltstone beds at site PN1. Intensities of initial NRM varied between 2.8×10^{-04} and 6.2×10^{-03} A m^{-1}. Generally, three components of magnetization were determined by the principal component analysis (Fig. 8d). The low-temperature component (A) was unblocked by 200 °C. The intermediate component (B) was generally demagnetized between 200 °C and 500 °C. The characteristic component (C) was isolated between 500 °C and 685 °C. Rock magnetic investigations suggest detrital haematite as carrier of the ChRM, with pigmentary haematite and goethite responsible for the acquisition of the B and A components, respectively.

The Northern Suture Zone

Shishi locality

We collected 24 orientated block samples from red-dark shale at two sites (SRS1 and SRS2). During the heating process, the colour of the specimens changed from dark to reddish at about 200 °C. No colour change was observed at such a low temperature in the samples from other localities. The initial NRM intensities varied between 5.2×10^{-03} and 1.53×10^{-02} A m^{-1}. Progressive thermal demagnetization revealed univectorial behaviour, which was generally broken down by 685 °C (Figs 9a and b). The palaeomagnetic behaviour here is consistent with the view of Turner (1980), who described the Type C magnetization of red beds as having a single, well-grouped direction which is stable to thermal demagnetization. Furthermore, she showed that the palaeomagnetic field tests for such magnetization often indicate a postfolding nature, with a direction close to the present Earth's field. The ChRM component obtained from the samples of the NSZ is thus considered to have been acquired in relatively recent time. Recent tectonic activity along the NSZ could be a possible cause of remagnetization. The complete overprinting at a relatively recent time may be due to the high proportion of unstable silicate minerals. Because of the postfolding remagnetization, both sites of the NSZ are regarded as unsuitable for Cretaceous palaeomagnetic conclusions.

PREVIOUS PALAEOMAGNETIC STUDIES IN CHITRAL

Klootwijk & Conaghan (1979) reported the first palaeomagnetic results from the belt of Devonian sedimentary rocks (pisolitic ironstones) along the Mastuj River (72.5°E, 36.2°N), Chitral district, northern Pakistan. The study area lies within the southern part of the Karakoram microplate, just north of the Reshun Fault. The ChRM direction obtained from the Devonian samples ($D = 318.1°$, $I = -6.4°$, $k = 14.0$, $\alpha_{95} = 16.7°$) gives an equatorial palaeoposition for the Chitral area. This palaeomagnetic direction was considered to be of primary origin and was interpreted in terms of a Late Devonian 'Laurasian' affinity of the Chitral region. Later, this controversial conclusion was refuted by the authors themselves, and the primary origin of the ChRM component was reinterpreted as a secondary remagnetization.

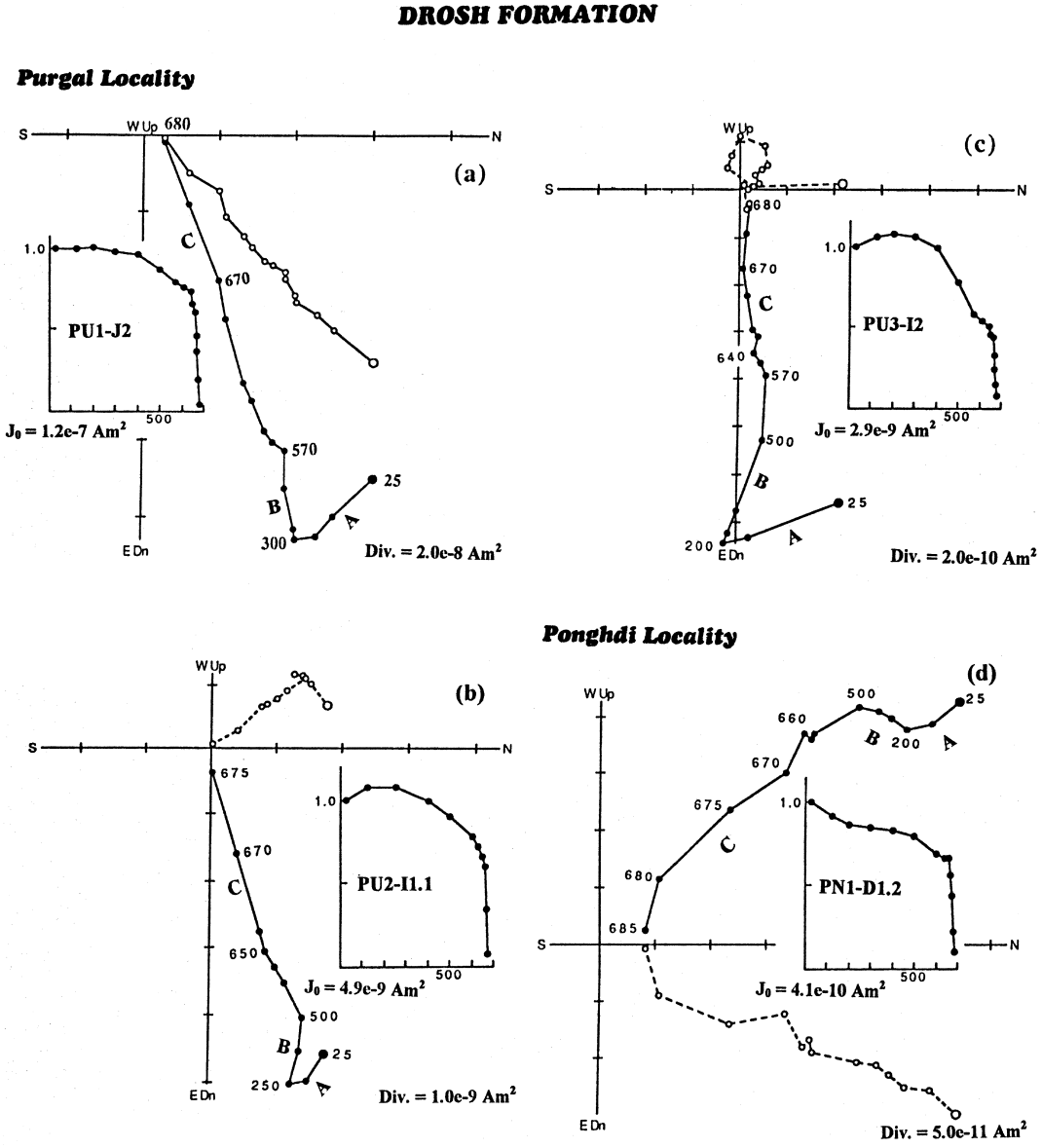


Figure 8. Results of the palaeomagnetic study for samples from the Drosh Formation at the Purgal and Ponghdi localities. (a), (b), (c), (d) Comments as for Fig. 5(b).

The second result was reported from the same area by Klootwijk *et al.* (1994). Thermal treatment of 333 block samples from Middle to Upper Devonian sediments, Permian quartz flysch, Permo–Triassic carbonates, and mid-Cretaceous red beds showed two interpretable components; that is, the recent field component (A) and the characteristic component (B). The high-temperature component (C) was only found in the samples of red mudstones. They interpreted the B component with equatorial NW–SE direction to be comparable to the observed characteristic component in the preliminary study on the Devonian pisolitic ironstones from the adjacent area. They further stated that the equatorial direction of this component is characteristic of a suite of secondary magnetizations observed both to the south and to the north of the Indus–Tsangpo Suture Zone, and also north of the western equivalent of this suture zone in northern Pakistan, which they attributed to initial India–Asia contact. On the basis of these results the authors suggested that the initial India–Asia contact was established at, or before, the KTB at equatorial to low-northern palaeolatitudes. The hard magnetization component (C) in the red mudstone was not used for palaeoreconstruction because of unclear bedding control. Tilt-corrected ChRM directions of the above two studies are given in Fig. 10.

DISCUSSION AND INTERPRETATION

From 169 red bed samples collected at 16 sites, about 119 were palaeomagnetically measured. A total of 75 reliable samples (104 specimens) from 12 sites were used for final palaeomagnetic statistics. The remaining 44 samples from all four formations were rejected on the following grounds.

(1) The ChRM components obtained from 16 samples of the sites SRS1 and SRS2 (NSZ) were found to be identical to the present Earth field direction (shown in Fig. 9 and listed in Table 3).

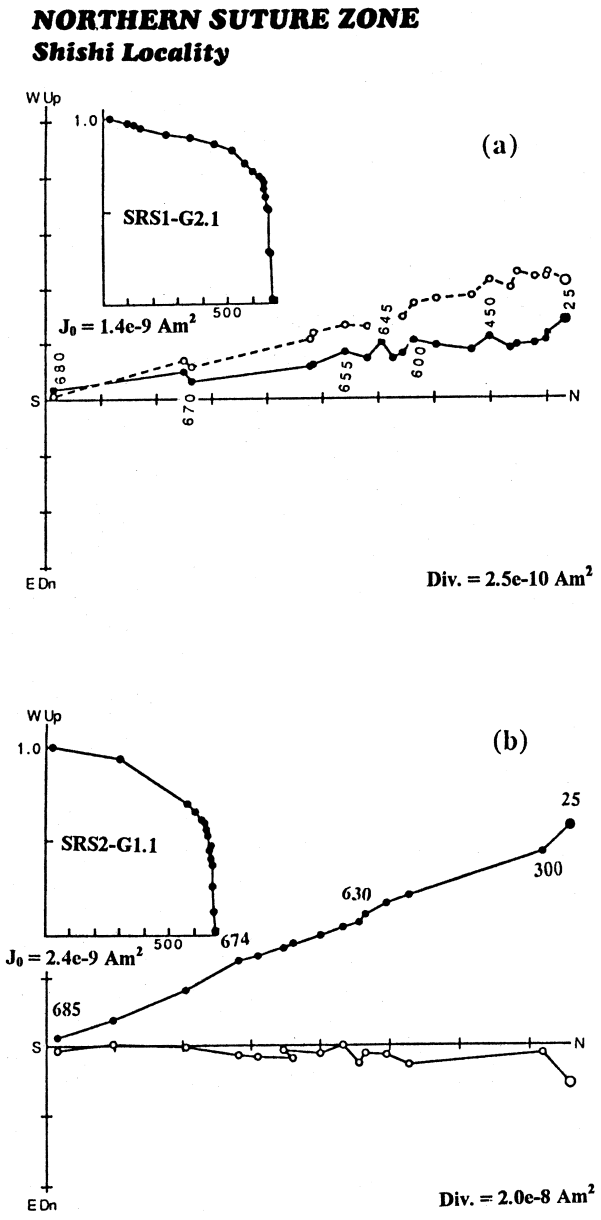


Figure 9. Results of the palaeomagnetic study for samples from the Northern Suture Zone at the Shishi locality. (a), (b) Comments as for Fig. 5(b).

(2) The ChRM component could not be reliably separated from 18 samples of the sites PN2 (PF) and TR1 (DF), because in the former site the samples appear very coarse grained and weathered, and in the latter the magnetite- and haematite-related components are possibly superimposed (not mentioned in the text and Tables).

(3) About 10 coarse-grained samples from the PF and RF sites had very noisy behaviour, and the components of magnetization could not be separated.

Thus on the basis of the palaeomagnetic results from 12 sites, three interpretable NRM components are separated from the red beds of the central and the southeastern Kohistan tectonic units. The low-temperature component (A) conforms to the present local field direction (Fig. 11). B is the most complex component and represents the secondary remagnetization, and the high-temperature component (C) can be regarded as the primary magnetization. The site-mean and formation-mean directions of the B and C components are listed in Tables 2 and 3, respectively. The pole positions and palaeolatitudes of the B and C components are listed in Table 4.

Intermediate component (B)

Because of the post-depositional diagenesis and prolonged tectonic events (especially after the India–Asia collision) in the study area, the separation of intermediate components in most samples was difficult. Using the method of best-fitting regression line, this uncertain intermediate component could only be separated from the samples of nine sites, which give a poorly defined mean direction (tilt-corrected: $D = 288.9°$, $I = 3.0°$, $k = 10.2$, $\alpha_{95} = 17.0°$) with west-northwesterly declination and very shallow positive inclination (Fig. 12). The precision parameter (k) increased slightly from 8.2 to 10.2 upon 100 per cent tilt correction (while taking into consideration the best possible result obtained after stepwise unfolding), indicating a prefolding or synfolding origin for this component. Comparison with the rock magnetic and optical microscopic results implies that pigmentary haematite is the carrier of the B component in the PF and RF samples, which was probably acquired as a result of post-depositional authigenic processes at, or slightly before, the India–Asia collision. The magnetite-related secondary magnetization in the DF samples was probably acquired by the movement of chemically active orogenic fluids at the time of the India–Asia collision. This view is supported by the temperature increase in excess of 300 °C deduced from fission-track resetting of zircon at 55–68 Ma from the adjacent area (Zeitler 1985). The similarity in directional behaviour of the above-mentioned secondary magnetizations (which reside in two separate magnetic mineral phases, i.e. haematite and magnetite) suggest that the authigenic production of pigmentary haematite was soon followed by the India–Asia collision. Although the reliability of the B component in the present study is questionable for some sites, the average tilt-corrected direction (Fig. 12) is in reasonable agreement with previously reported ChRM directions from the eastern Hindukush mountains (Klootwijk & Conaghan 1979; Klootwijk *et al.* 1994). The declination difference of 25°–30° between present and previous studies is probably due to the differences in rotation across the NSZ, where in the former case most of the sampling sites are situated on south side of the suture zone, and in the latter exclusively to the north side. The equatorial to low northern palaeolatitude obtained here for the B component supports the interpretation of Klootwijk *et al.* (1991, 1994), that initial India–Asia contact was established at, or before, the KTB at equatorial to low northern palaeolatitudes. The counter-clockwise rotation obtained from the B component (relative to Eurasian and Indian apparent polar wander paths, APWPs, of Besse & Courtillot 1991) corresponds well to the regional superstructure of the Hindukush–Pamir–Karakoram syntaxial bend.

Characteristic component (C)

The rock magnetic and optical microscopic results suggest that the ChRM in the PF and RF samples is carried by a high-temperature and high-coercivity detrital haematite. High-temperature but relatively low-coercivity haematite is the

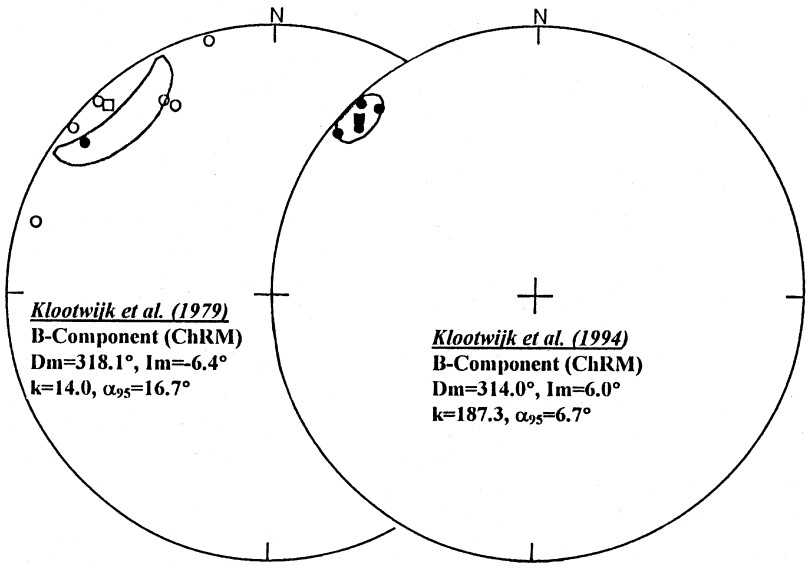


Figure 10. Equal-area projections of the sample-mean and formation-mean directions for the B component (ChRM) from previous palaeomagnetic results in the study area (Klootwijk *et al.* 1979, 1994). Closed (open) symbols are downward (upward) directions.

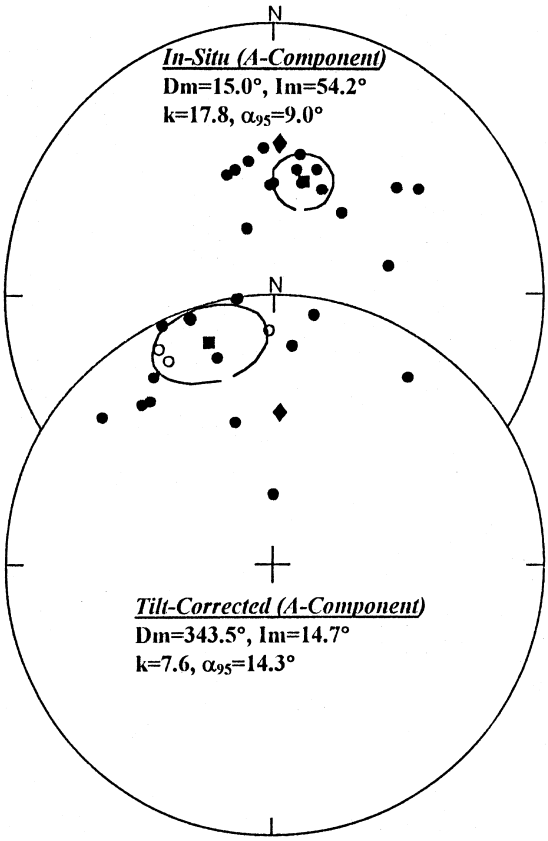


Figure 11. Equal-area projections of the formation-mean direction for the A component (16 site-mean directions from the Reshun, Purit, and Drosh formations as well as the Northern Suture Zone) in both geographical and stratigraphic coordinates. Diamond symbols indicate the present local magnetic field direction of the Earth. The *in situ* mean direction of the most unstable A component is almost identical to the present field direction of the study area. Closed (open) symbols are downward (upward) directions.

carrier of ChRM in the DF samples. The red beds of the DF are interbedded as thin bands within the volcanic andesites. These red beds are probably derived from the volcanic ashes and other fine sediments of the surrounding igneous rocks, which could result in the presence of very fine-grained magnetic minerals. Turner (1980) stated that in some cases the original sediment may have been rich in magnetite, but after progressive diagenesis (including magnetite oxidation), the NRM related to very fine magnetite could be replaced by a haematite-related chemical remanent magnetization (CRM). The haematite grains in the DF samples are therefore considered to be formed by martitization or hydrothermal alteration of the pre-existing detrital magnetite, suggesting the ChRM origin as a CRM. Butler (1992) has stated that CRM in fine-grained sediments might be acquired more quickly than in coarse-grained sediments. The arguments for a quickly acquired CRM in the DF samples are supported by the well-defined character of the ChRM direction and the limited site-mean dispersion.

The site-mean directions of the ChRM component can be divided into two groups (Figs 13a,b and c). As shown in Fig. 13(a), the formation-mean direction of the seven PF sites shows a northwestward declination and very shallow negative inclination ($D = 311.2^\circ$, $I = -4.4^\circ$, $k = 12.4$, $\alpha_{95} = 17.8^\circ$). The precision parameter (k) increases from 5.5 to 12.4 upon tilt correction, indicating a positive fold test. The dual polarity, the high coercivity and unblocking temperatures, and the positive fold and conglomerate tests indicate that the ChRM in the PF red beds was acquired as detrital remanent magnetization (DRM) before the folding episode of the India–Asia collision. As shown in Fig. 14, the mean declination obtained from the PF sites group indicates a counter-clockwise rotation of about 66° and 15° with respect to Eurasia and India, respectively (using the Cretaceous APWPs of Besse & Courtillot 1991). Tapponier & Molnar (1977) have argued that, subsequent to the India–Eurasia collision, much of central Afghanistan was extruded to the west between the Chaman and Herat faults, and that Tibet was squeezed out to the east along the

Table 4. Pole positions and palaeolatitudes for the B and C components in stratigraphic coordinates.

Component	*Formation*	*Sites Used*	*n*	*Long.*	*Lat.*	d_p	d_m	*Plat.*
B (?)	RF, PF, and DF	BRS1, PRS1, PRS2, NS3, MI2, MI3, PU1, PU2, PU3	9	134.7°E	16.3°S	8.5°	17.0°	1.5°N
C	PF	PRS1, PRS2, NS1, NS2, NS3, MI2, MI3	7	132.5°E	31.1°S	8.9°	17.8°	2.2°S
C	RF and DF	BRS1, PU1, PU2, PU3, PN1	5	167.0°E	5.0°N	10.6°	5.3°	1.1°N

n = number of sites used for pole position calculation; Long. = pole longitude; Lat. = pole latitude; d_p,d_m = oval of 95% confidence about the pole position; Plat. = palaeolatitude.

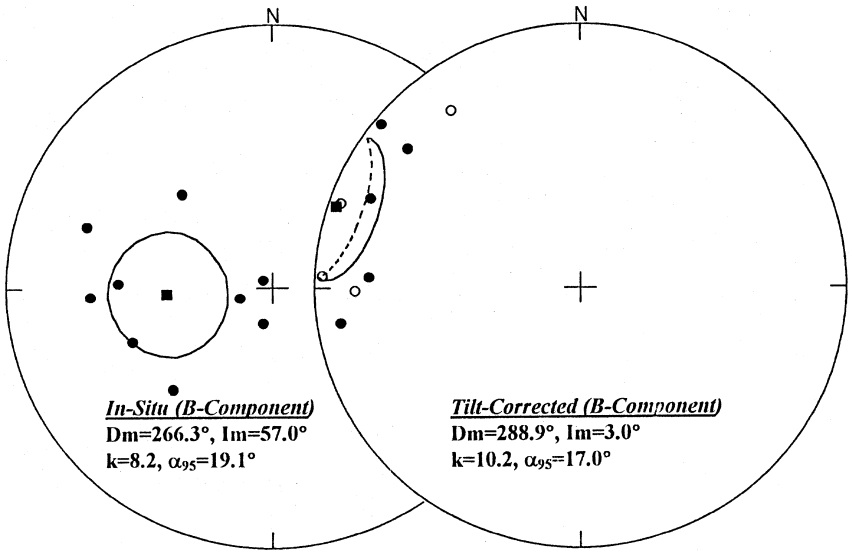


Figure 12. Equal-area projections of the formation-mean direction for the B component (nine site-mean directions from the Reshun, Purit, and Drosh formations) in both geographical and stratigraphic coordinates. Closed (open) symbols are downward (upward) directions.

Karakoram and Altyn Tagh faults. The counter-clockwise palaeomagnetic rotation obtained from the PF red beds (located in the northwestern part of the syntaxis) is consistent with the superstructure of the Hindukush–Pamir–Karakoram bending.

The formation-mean direction from the RF and DF (five sites) can be interpreted in two ways (Figs 13b and c). First, the results can be interpreted to be of normal polarity with northeastward declination and very shallow negative inclination (tilt-corrected direction: $D = 83°$, $I = -2.2°$, $k = 52.7$, $\alpha_{95} = 10.6°$). Second, a reverse polarity interpretation leads to an antipodal direction with southwestward declination and very shallow positive inclination (tilt-corrected direction: $D = 263.0°$, $I = 2.2°$, $k = 52.7$, $\alpha_{95} = 10.6°$). The latter interpretation seems to be more realistic than the former one, because the declination value is consistent with the regional structure of the study area. The precision parameter (k) increases from 37.1 to 52.7 (positive fold test), suggesting that the ChRM was acquired before the folding episode of the India–Asia collision. On the basis of Cretaceous APWPs (Besse & Courtillot 1991), the northwestward declination indicates a counter-clockwise rotation of about 114° and 63° relative to Eurasia and India, respectively (Fig. 14), which is also consistent with the regional superstructure. The difference in rotation between the PF and DF + RF is probably due to the fact that the rocks of these formations acted as dismembered thrust sheets after the ChRM acquisition.

Palaeolatitude and its application to palaeogeography

The shallow inclinations obtained from the PF and DF + RF indicate that the ChRMs were acquired at equatorial to low southern and low northern palaeolatitudes, respectively. Comparison of the inclination values with the northward motion of the study area imply that the ChRM of the PF red beds was acquired in the mid-Cretaceous at equatorial to low southern palaeolatitudes, while those related to the RF and DF red beds were probably acquired in the Late Cretaceous at equatorial to very low northern palaeolatitudes. Subsequently, the secondary magnetization components (B) were probably

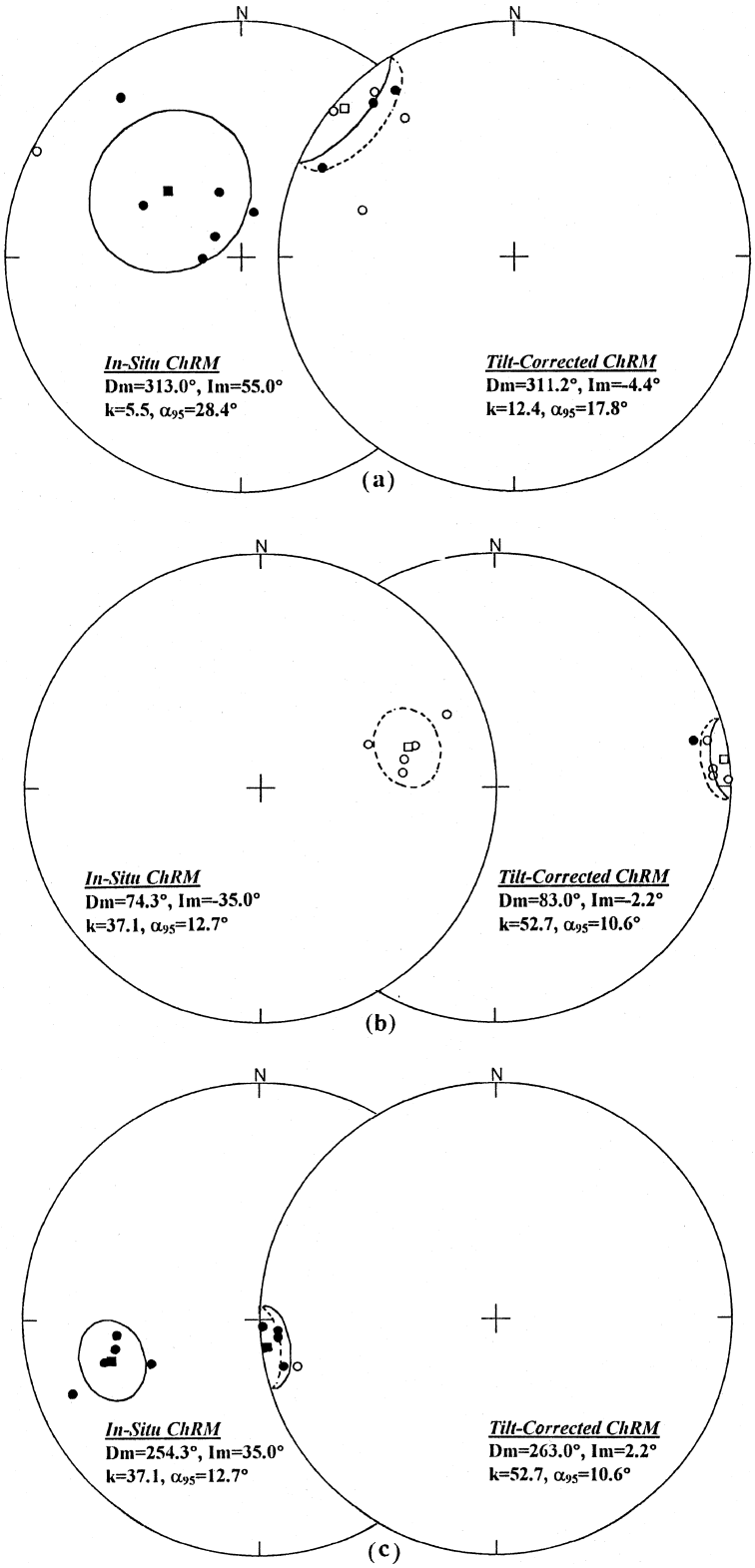


Figure 13. Equal-area projections of the formation-mean directions for the ChRM components in both geographical and stratigraphic coordinates. Closed (open) symbols are downward (upward) directions. (a) Seven sites of the Purit Formation. (b) Five sites of the Reshun and Drosh formations. (c) Antipodal directions of (b).

acquired at equatorial to low northern palaeolatitudes by chemical and thermo-chemical processes. The Cretaceous palaeolatitudes obtained from the ChRM inclinations are 2.2°S and 1.1°N for the PF and RF + DF, respectively. These equatorial palaeolatitudes are in good agreement with the expected palaeogeographical positions of the Kohistan–Karakoram composite unit in the mid- to Late Cretaceous. This was the area where the red bed sediments were deposited

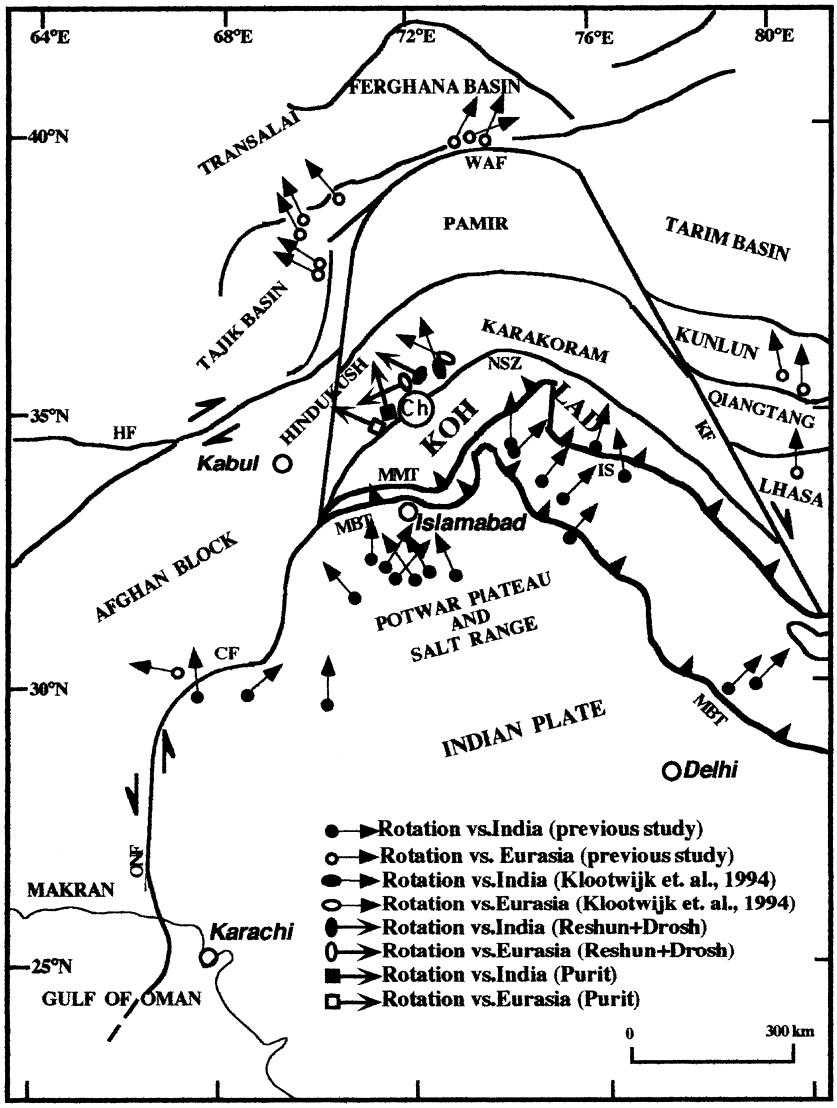


Figure 14. Regional tectonic map of south central Asia, showing sutures, major faults and blocks (modified from Stöcklin 1977). The sampling area is marked by Ch within a circle, which is located across the Northern Suture Zone. Magnitude and sense of rotation within the western syntaxis of the Himalaya are marked by arrows (Besse & Courtillot 1991; Klootwijk *et al.* 1991, 1994; Appel *et al.* 1995). The rotation deduced from the ChRM components (this study) is counter-clockwise for both the Purit and Reshun + Drosh formations. KOH: Kohistan, LAD: Ladakh, WAF: Wanch-Akbaytal Fault, HF: Heart Fault, KF: Karakoram Fault, NSZ: Northern Suture Zone, MMT: Main Mantle Thrust, IS: Indus Suture, MBT: Main Boundary Thrust, CF: Chaman Fault, ONF: Ornach-Nal Fault.

in a warm and humid environment after the arc-continent collision along the NSZ, which was formed after the consumption of a back-arc basin at 102–85 Ma (Pudsey *et al.* 1985; Treloar *et al.* 1989). Although not very significant within the α_{95} level, from the palaeolatitudinal difference between the B and C components (from the KTB to mid-Cretaceous) it is roughly estimated that the study area was pushed northwards about 4° in the time between the deposition of the red beds and the India–Asia collision. A northward displacement of the sampling area of about 400 km between the acquisition of the B and C components is acceptable in the sense that during the said period the oceanic part of the subducting Indian plate was pushing the Kohistan–Karakoram composite unit northwards. Comparison of the mid-Cretaceous palaeoposition of the Kohistan–Karakoram composite unit (~2.2°S) with the present-day latitude (~36°N) indicates about 4000 km of northward displacement, about 90 per cent of which postdates the acquisition of the B component at or slightly before the India–Asia collision (Fig. 15). The displaced lithosphere was probably accommodated by continental underthrusting, intercontinental thrusting, internal deformation, and lateral extrusion along major and minor faults (Patriat & Achache 1984; Molnar & Tapponier 1975). Comparison of the palaeolatitudinal positions for the Kohistan–Karakoram composite unit obtained here with the Cretaceous configuration of Asia (Chen *et al.* 1993b) reveals the existence of a wide Neo-Tethyan oceanic gap (Fig. 15). Such a wide gap is not compatible with geological interpretations, which predict that the Cretaceous Kohistan–Ladakh arc was not far from the southern margin of Asia (Pudsey 1986; Treloar *et al.* 1989). More palaeomagnetic data of Cretaceous age from the South Asian margin (particularly the Afghan and Karakoram blocks) are needed

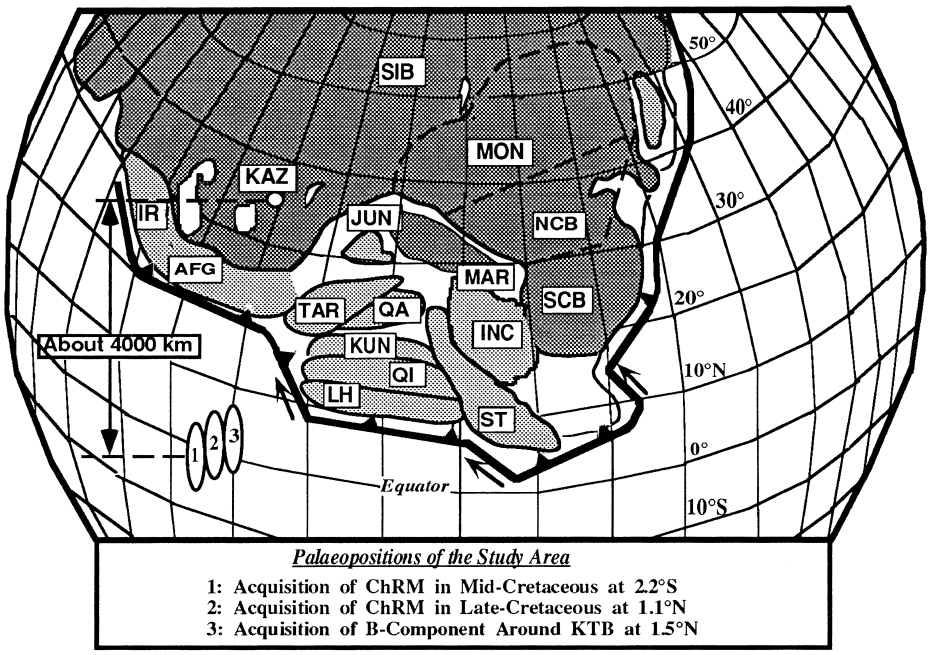


Figure 15. Comparison of the palaeolatitudes obtained from the intermediate B component and the characteristic C components with the Cretaceous palaeogeographical reconstruction of Asia (Chen *et al.* 1993b). Comparison with the present-day latitude of the study area (35°–36N) implies that about 4000 km of crustal shortening has taken place since the mid-Cretaceous ChRM acquisition at equatorial palaeolatitudes, about 90 per cent of which post-dates the acquisition of the B component. AFG: Afghanistan, INC: Indochina, IR: Iran, JUN: Junggar, KAZ: Kazakhstan, KUN: Kunlun, LH: Lhasa, MAR: Makram, MON: Mongolia, NCB: North China Block, QA: Qaidam, OI: Qiangtang, SIB: Siberia, SCB: South China Block, ST: Shan Thai, TAR: Tarim.

to reconstruct fully the palaeopositions of microcontinental blocks which were separated from the northern margin of Gondwana and amalgamated around the northwestern Himalayan syntaxis. However, on the basis of geological information and the ChRM directions detailed here, we present a simple model for the Kohistan–Karakoram composite unit in mid-Cretaceous time, shown in Fig. 16. This cartoon shows an equatorial palaeolatitude for the sampling area at the time of ChRM acquisition. The cartoon further implies that, in the mid-Cretaceous, the Kohistan–Karakoram composite unit formed part of the South Asian margin. Another important issue is the position of the NSZ during the Cretaceous. The Late Cretaceous inclination values across the NSZ are almost identical, which indicates that suturing was complete at the time of the ChRM acquisition. This is in good agreement with the ^{39}Ar–^{40}Ar and K–Ar results from North Pakistan, which date the completion of suturing at 102–85 Ma (Treloar *et al.* 1989). According to Stöcklin's (1977) interpretation, most of the Gondwanic microcontinents became sutured to the stable Eurasia in pre-Cretaceous time. Thus the Cretaceous origin of the ChRM components (this study) does not allow a distinction between a Eurasian and a Gondwanan affinity for the eastern Hindukush region of the Karakoram microplate.

CONCLUSIONS

The dominant magnetic mineral identified in the samples of the PF and RF is a detrital specular haematite, while a relatively low-coercivity haematite (probably chemical in origin) is the main magnetic carrier in the DF samples.

Magnetizations related to the B component were probably acquired at, or slightly before, the India–Asia collision, and are thus considered to be thermochemical and chemical in origin. The ChRM component in the PF red beds was acquired as a DRM in mid-Cretaceous time, while those in the RF and DF red beds were probably acquired in the Late Cretaceous as a DRM and CRM, respectively.

Palaeomagnetic declinations of the ChRM components indicate that the studied thrust sheets experienced a counter-clockwise rotation of 66°–114° and 15°–63° relative to Eurasia and India, respectively, which is consistent with the regional superstructure of the Hindukush–Pamir–Karakoram syntaxial bend.

Palaeolatitudes obtained from the ChRM inclinations are about 2.2°S for the PF and 1.1°N for the DF + RF, while that for the B component is about 1.5°N(?). These palaeolatitudes imply about 4000 km of northward displacement since the mid-Cretaceous, of which about 90 per cent post-dates the terminal collision of India with Asia

Nearly identical Late Cretaceous inclinations across the NSZ indicate that suturing was complete at the time of the ChRM acquisition.

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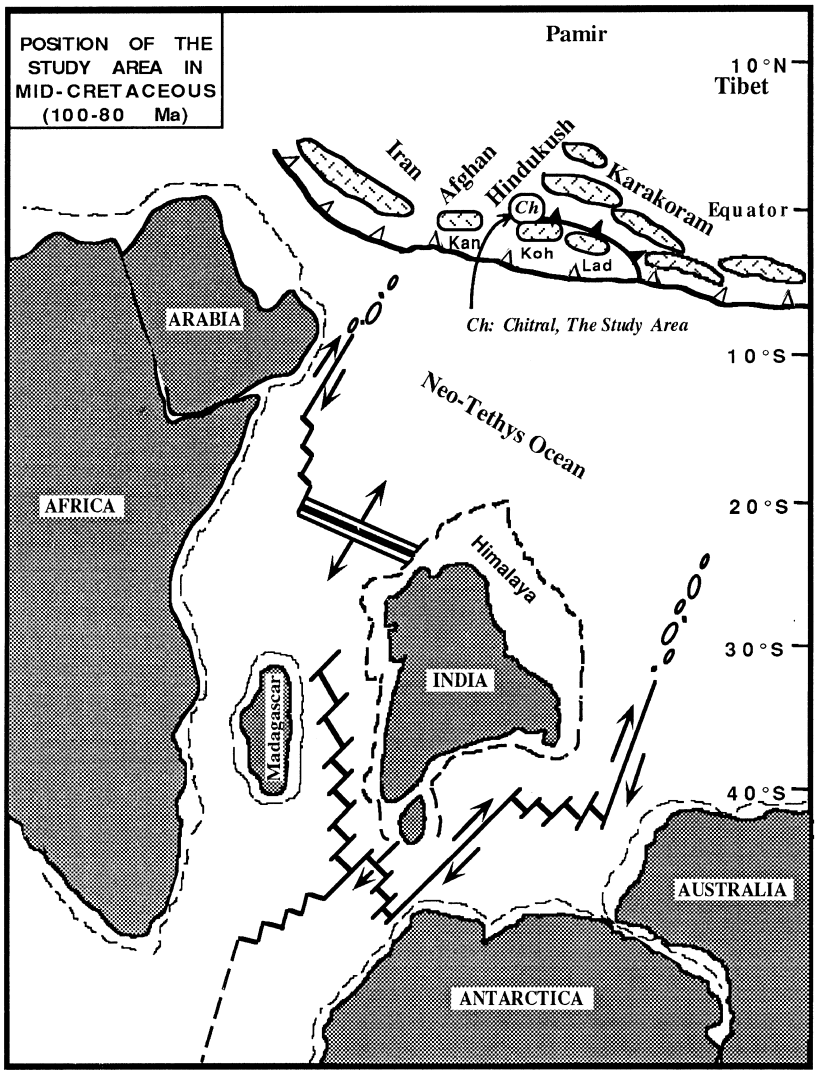


Figure 16. Reconstruction showing the palaeoposition of the Kohistan–Karakoram composite unit in the mid-Cretaceous (100–80 Ma). At that time, the Indian shield was still positioned at moderate southern latitude across the Neo-Tethys Ocean. The study area is marked by Ch within an oval. Kan: Kandahar, Koh: Kohistan, Lad: Ladakh.

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